Sulabham

APPSC AEE 2026 · Module 3: Surveying

Paper 3 Civil Engineering · Post Codes 1, 3, 4 and 7 · Study week 7 · English medium

How to use this module

1. **Understand:** read the topics in order — basic idea → why → how → analogy → traps.
2. **Connect:** study the photo and the diagram, then the "Connections" line.
3. **Remember:** learn the memory code, close the page and answer the self-recall questions aloud.
4. **Apply:** take the 5-question quiz at the end of each topic. Tap an option; the answer and the reason for every option appear only after you choose.
5. At the end, write the Notebook page from memory, then take the Module 3 test.

Week plan (Week 7): Day 1 → Topics 1–2 · Day 2 → Topic 3 · Day 3 → Topic 4 · Day 4 → Topics 5–6 · Day 5 → Topics 7–8 · Day 6 → Topics 9–10 · Day 7 → Notebook page + test. No calculator is allowed in the exam.

How Surveying was asked in APPSC AEE 2023 (Paper III Civil, 22 Aug 2023, Shift 2)

Evidence (counted from the official paper and key)
Surveying had 11 of the 150 questions (Q131 to Q141). Only this one shift was available to us, so topics with a zero below were simply not picked in this paper.

Topic	Questions	Share
Topic 1: Fundamentals, scales, errors	1	
Topic 2: Chain and tape	2	
Topic 3: Compass	2	
Topic 4: Theodolite and traversing	0	
Topic 5: Levelling	1	
Topic 6: Contours and DEM	0	
Topic 7: Tacheometry, EDM, total station	1	
Topic 8: Triangulation and adjustment	0	
Topic 9: Plane table, photogrammetry	2	
Topic 10: Remote sensing, GPS, astronomy	2	

How the 11 questions were framed	Count	Question numbers
Direct one-line concept	4	Q134, Q138, Q140, Q141
Match the lists (P, Q, R, S)	2	Q132, Q135
Statement with two conclusions	2	Q133, Q139
Numerical	2	Q136 (with figure), Q137
Assertion–Reason	1	Q131

Our analysis (not a prediction)
Three of the eleven questions (Q131, Q133, Q139) were built by **reversing or miscalculating a true statement**, with the key "both false" or "neither correct". So read every statement word by word and work the numbers before agreeing. Photogrammetry and remote sensing together gave four questions — as many as chain and compass together — so do not leave the modern topics for the end.

Topic 1: Fundamentals — Principles, Classification, Scales and Errors

Syllabus line: *Classification of surveys, scales, accuracy.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 1 question

- Assertion–Reason on the principle of surveying: the paper wrote 'working from **part to whole**', which is wrong — the principle is whole to part, and working from part to whole magnifies errors. Key: both A and R are false

(Q131).

1. Basic idea

Surveying is the art of finding the relative positions of points on, above or below the earth's surface by measuring distances, directions and elevations, and then showing them on a plan or map. A **plan** is drawn to a large scale (a building site); a **map** is drawn to a small scale (a district).

2. Why it is needed

No road, canal, building, pipeline or dam can be designed until the engineer knows the shape of the ground. Surveying gives that picture before construction, and sets out the design on the ground during construction.

3. How it works

Two principles govern every survey:

1. **Work from the whole to the part.** First fix a few widely spaced control points very accurately; then fill in the details with less precise work. Errors stay small and local, and do not pile up.
2. **Fix every new point by at least two measurements** from two known reference points (two distances, or a distance and an angle, or two angles).

Basis	Types	Remember
Curvature of the earth	**Plane surveying** — earth treated as flat. **Geodetic surveying** — curvature is taken into account	Plane survey is used for areas up to about 250 km²; an arc of 12 km is only about 1 cm longer than its chord
Nature of the field	Land (topographic, cadastral, city, engineering), marine or hydrographic, astronomical	Cadastral = property boundaries (revenue maps)
Purpose	Engineering, military, mine, geological, archaeological	Reconnaissance → preliminary → location surveys
Instrument used	Chain, compass, theodolite, level, plane table, tacheometric, photogrammetric, EDM / total station, GPS	This is how the syllabus topics are arranged
Method	Triangulation, traversing	Topic 4 and Topic 8

Scales

- **Representative fraction (RF)** = map distance ÷ ground distance, both in the same unit. 1 cm = 50 m → RF = 1/5000.
- **Plain scale** reads two dimensions (metres and decimetres). **Diagonal scale** reads three (metres, decimetres, centimetres). **Vernier scale** reads a fraction of the smallest main division. **Scale of chords** sets out angles.
- **Least count of a vernier** = smallest main-scale division ÷ number of vernier divisions. Direct vernier: n vernier divisions = (n − 1) main divisions.
- **Shrunk map:** shrinkage factor = shrunk length ÷ original length; shrunk scale RF = shrinkage factor × original RF.
- **Measured with a wrong scale:** correct length = (RF of wrong scale ÷ RF of correct scale) × measured length; for areas use the square of that ratio.

Kind of error	Cause	Behaviour
Mistake (blunder)	Carelessness — misreading, wrong booking	No rule; found only by checking
Systematic (cumulative)	Known cause — wrong chain length, temperature, sag	Same sign every time; grows in proportion to the length L; removed by a correction formula
Accidental (random, compensating)	Unknown small causes	Sometimes +, sometimes −; grows only as √L; follows the law of probability

Accuracy means closeness to the true value; **precision** means closeness of repeated readings to each other. A wrongly calibrated tape can be precise yet inaccurate.

4. Everyday analogy

Think of a jigsaw puzzle. If you first fix the four corners and the border (whole) and then fill the inside (part), small misfits stay where they are. If you start from one inner piece and grow outward, a small tilt keeps growing until the

border does not close.

5. Confusing areas and exam traps

- **Whole to part**, never part to whole — the 2023 paper tested exactly this reversal.
- Cumulative error ∝ L; compensating error ∝ √L.
- A larger RF denominator means a **smaller** scale (1:50,000 is smaller than 1:5,000).
- Diagonal scale works on similar triangles; it does not need a vernier.

6. Connections

Errors and their adjustment return in Topic 8. Scale returns in aerial photographs (Topic 9). Reconnaissance → preliminary → final location is also the order of highway surveys (2023 Q142, Transport module).

Memory code: "Whole first, Two ties"

Whole first	Control points for the whole area first, details later — errors stay local
Two ties	Every new point is tied by at least two measurements from known points
C-L, R-√L	Cumulative error ∝ L; Random (compensating) error ∝ √L

Self-recall (close the notes, answer aloud, then check)

1. State the two principles of surveying.
   Work from whole to part; locate each new point by at least two measurements from fixed reference points.
2. A map has RF 1/25,000. What ground distance does 6 cm show?
   6 × 25,000 cm = 1,50,000 cm = 1.5 km.
3. How do cumulative and compensating errors grow with length?
   Cumulative in proportion to L; compensating in proportion to √L.

Quick check: 5 questions

1. **Assertion (A):** In surveying, the main control points are fixed first and the details are filled in afterwards.
Reason (R): Working from whole to part localises errors and prevents their accumulation.

   1. Both A and R are true, and R is the correct explanation of A
   2. Both A and R are true, but R is NOT the correct explanation of A
   3. A is true, but R is false
   4. A is false, but R is true

   1. Correct. Both are true and R is the reason for A — this is the first principle of surveying.
   2. R is the direct reason for the procedure in A.
   3. R is true.
   4. A is true.

2. On a map of RF 1 : 50,000, two towns are 4 cm apart. The ground distance between them is

   1. 0.2 km
   2. 20 km
   3. 200 km
   4. 2 km

   1. Ten times too small.
   2. Ten times too large.
   3. A hundred times too large.
   4. Correct. 4 × 50,000 cm = 2,00,000 cm = 2 km.

3. A plan drawn to a scale of 1 cm = 50 m was measured by mistake with a scale of 1 cm = 40 m, and a line was recorded as 320 m. The true length of the line is

   1. 256 m
   2. 320 m
   3. 400 m
   4. 500 m

   1. Multiplies by 40/50 — the ratio is upside down.
   2. This is the wrong reading itself.
   3. Correct. Length on plan = 320/40 = 8 cm; true length = 8 × 50 = 400 m.
   4. Would need 10 cm on the plan.

4. Consider the following statements about errors in surveying:
   I. Systematic (cumulative) errors are proportional to the length measured.
   II. Accidental errors are proportional to the square root of the length measured.
   III. Mistakes can be removed by applying a correction formula.
   IV. Accidental errors follow the law of probability.
   Which of the above statements are correct?

   1. I and II only
   2. I, II and IV only
   3. III and IV only
   4. I, II, III and IV

   1. Leaves out IV, which is true.
   2. Correct. I, II and IV are true. Mistakes (III) follow no rule and are found only by checks.
   3. III is false.
   4. Includes the false statement III.

5. The smallest division on the main scale of a theodolite is 20′, and 60 vernier divisions coincide with 59 main-scale divisions. The least count is

   1. 20″
   2. 10″
   3. 30″
   4. 1′

   1. Correct. Least count = 20′/60 = 20″.
   2. Would need 120 vernier divisions.
   3. Would need 40 vernier divisions.
   4. Would need 20 vernier divisions.

Topic 2: Measurement of Distances — Chain and Tape

Syllabus line: *Measurement of distances — direct methods.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 2 questions

- Length across a river by similar triangles on a continuing chain line: PQ = 45 m (Q136, figure-based).
- Error that can be classed as a compensating error: **variation in pull** (Q140).

1. Basic idea

In **chain surveying** only lengths are measured — no angles. The area is divided into a framework of triangles, because a triangle is the only figure that can be plotted from its sides alone.

2. Why it is used

It is the simplest and cheapest method. It suits small, fairly level, open areas with simple details. It is not suitable for large areas, crowded towns or hilly ground.

3. How it works

Chain / tape	Length	Note
Metric chain	20 m (100 links) or 30 m (150 links)	One link = 20 cm; brass tallies at every 5 m
Gunter's (surveyor's) chain	66 ft, 100 links	10 square chains = 1 acre; 80 chains = 1 mile
Engineer's chain	100 ft, 100 links	One link = 1 ft
Revenue chain	33 ft, 16 links	Cadastral work
Steel tape	30 m or 50 m	Standard work
Invar tape	Alloy of nickel (36 %) and steel	Very low thermal expansion — base-line measurement

Lines of the framework: base line — the longest line, the backbone of the survey; **check (proof) line** — proves the accuracy of the framework; **tie line** — joins points on main lines to pick up interior details; **offset** — a short lateral measurement from the chain line to a detail. Triangles should be **well-conditioned**: no angle less than 30° or more than 120°.

Instruments for right angles: cross staff, **optical square** (two mirrors at 45°, works on double reflection) and prism square. **Ranging** means placing intermediate points on a straight line: direct ranging when the ends are intervisible,

indirect (reciprocal) ranging when a hill hides them.

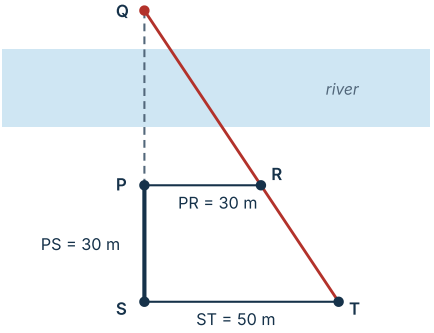
Wrong length of chain

- True length = (L′ / L) × measured length, where L′ = actual length of the chain and L = its designated length.
- True area = $(L' / L)^2$ × measured area; true volume = $(L' / L)^3$ × measured volume.
- Chain **too long** → it fits fewer times → measured distance is **too small** → correction is **positive**. Chain too short → correction negative.

Tape corrections

- Temperature: $C_t = \alpha (T_m - T_0) L$ — positive when the field temperature is above the standard.
- Pull: $C_p = (P - P_0) L / (A E)$ — positive when the applied pull is more than the standard pull.
- Sag: $C_s = W^2 L / (24 P^2)$, where W = weight of the tape between supports — **always negative**.
- Slope: $C = h^2 / (2L)$, or $L (1 - \cos \theta)$ — **always negative**.
- Reduction to mean sea level: $C = L h / R$ — negative for points above sea level.
- **Normal tension** is the pull at which the pull correction just cancels the sag correction.

Type of error	Examples in chaining
Cumulative, positive (measured length too long)	Chain too short, bad ranging, sag, slope not allowed for, chain not straight, low temperature
Cumulative, negative (measured length too short)	Chain too long, high temperature, pull more than standard
Compensating	Variation in pull, careless marking of chain ends, reading fractions of a link



Triangles QPR and QST are similar

PQ / QS = PR / ST

PQ / (PQ + 30) = 30 / 50

50 PQ = 30 PQ + 900

PQ = 45 m (2023 Q136)

PR and ST are both perpendicular to the chain line, so the two triangles have equal angles.

Obstacle to chaining but not to vision (a river): set out two perpendiculars on the near bank and use similar triangles to get the unmeasurable length.

Obstacle	Example	Method
Chaining free, vision obstructed	Rising ground or a hill between the stations	Reciprocal (indirect) ranging
Chaining obstructed, vision free	River, pond	Similar or right-angled triangles on the bank
Both obstructed	Building	Run a parallel line by equal perpendicular offsets

4. Everyday analogy

A tailor's tape that has stretched reads less than the true cloth length, so the customer gets extra cloth. In the same way a chain that is too long gives a measured distance that is too small, and the correction must be added.

5. Confusing areas and exam traps

- Chain too long → measured value too small → **add**. Students often reverse this.
- Sag and slope corrections are always negative, whatever the conditions.
- In the sag formula W is the total weight of the tape between supports; with weight per metre w, $C_s = L (wL)^2 / 24P^2$.
- Variation in pull is compensating; a pull that is always higher than standard is cumulative.

6. Connections

Well-conditioned triangles return in triangulation (Topic 8). Slope and sea-level reductions are applied automatically by a total station (Topic 7).

Memory code: "Long chain, Less reading, Add"

Long → Less → Add	Chain too long → measured length less → positive correction
S-S negative	Sag and Slope corrections are always negative
30–120	Angles of a well-conditioned triangle lie between 30° and 120°

Self-recall (close the notes, answer aloud, then check)

1. Why is the area divided into triangles in chain surveying?
   A triangle is the only figure that can be plotted from the lengths of its sides alone, without any angle.
2. Write the sag correction formula and its sign.
   $C = W^2L/(24P^2)$, always negative (W = weight of tape between supports, P = pull).
3. Name one compensating error in chaining.
   Variation in pull (also careless marking, estimating fractions of a link).

Quick check: 5 questions

1. A 20 m chain was found to be 10 cm too long after measuring a line as 1500 m. The true length of the line is

   1. 1492.5 m
   2. 1500.0 m
   3. 1507.5 m
   4. 1515.0 m

   1. Applies the correction with the wrong sign.
   2. Ignores the error in the chain.
   3. Correct. True length = (20.1/20) × 1500 = 1507.5 m — a chain that is too long gives a positive correction.
   4. Doubles the correction.

2. A distance of 100 m is measured along a uniform slope between two points whose difference in level is 4 m. The horizontal distance is nearly

   1. 99.84 m
   2. 99.92 m
   3. 99.20 m
   4. 100.08 m

   1. Uses h^2/L instead of $h^2/2L$.
   2. Correct. Slope correction = $h^2/2L$ = 16/200 = 0.08 m (negative) → 99.92 m.
   3. Ten times the correction.
   4. Slope correction is never positive.

3. Consider the following statements about tape corrections:

   I. The sag correction is always negative.
   II. The slope correction is always negative.
   III. The temperature correction is always positive.
   IV. The pull correction may be positive or negative.

   Which of the above statements are correct?

   1. I and II only
   2. I, II and III only
   3. I, II and IV only
   4. I, II, III and IV

   1. Leaves out IV, which is true.
   2. III is false.
   3. Correct. Temperature correction (III) is negative when the field temperature is below the standard.
   4. Includes the false statement III.

4. To continue a chain line AB across a river (B on the far bank), a perpendicular AC = 40 m is set out at A. A line at right angles to CB at C meets the chain line produced backwards at D, and AD = 20 m. The length AB is

1. 20 m
2. 40 m
3. 160 m
4. 80 m

1. That is AD.
2. That is AC.
3. Doubles the answer.
4. Correct. Triangles ABC and ACD are similar: AB/AC = AC/AD → AB = AC^2/AD = 1600/20 = 80 m.

5. Match List-I with List-II:

List-I	List-II
A. Optical square	1. 66 ft long with 100 links
B. Invar tape	2. Setting out right angles
C. Ranging rod	3. Base-line measurement
D. Gunter's chain	4. Marking intermediate points on a line

1. A-2, B-3, C-4, D-1
2. A-3, B-2, C-4, D-1
3. A-2, B-4, C-3, D-1
4. A-4, B-3, C-1, D-2

1. Correct. Optical square → right angles (double reflection); invar tape → base lines; ranging rod → marking points on a line; Gunter's chain → 66 ft with 100 links.
2. Swaps the optical square and the invar tape.
3. Swaps the invar tape and the ranging rod.
4. Gunter's chain is the one of 66 ft.

Topic 3: Measurement of Directions — Compass Surveying

Syllabus line: *Measurement of directions, prismatic compass, local attraction.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 2 questions

- The zero/360° graduation of a prismatic compass is marked at the **south end** (Q134).
- For a whole circle bearing between 270° and 360°, reduced bearing = 360° − WCB (Q141).

1. Basic idea

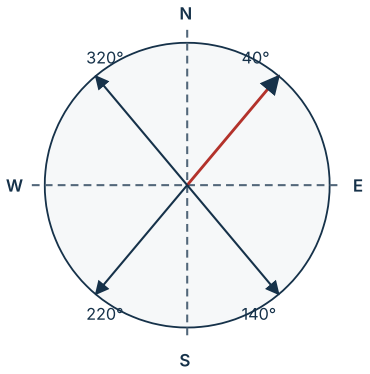
A chain gives only lengths. To plot a line we also need its **direction**. The direction of a line is given by its **bearing** — the horizontal angle it makes with a reference direction called a **meridian**.

2. Why a compass

A freely suspended magnetic needle always settles along the **magnetic meridian**. So a compass gives the bearing of every line independently; an error in one line is not carried to the next. It suits traverses along roads, rivers and forest boundaries.

3. How it works

Meridian	Meaning
True meridian	Line through the point and the geographic north and south poles; fixed
Magnetic meridian	Direction shown by a free magnetic needle; changes with place and time
Grid meridian	Central true meridian adopted for a whole map sheet
Arbitrary meridian	Any convenient direction, for example towards a tower



WCB → reduced bearing (RB)

0°–90° (NE): RB = WCB → N 40° E

90°–180° (SE): RB = 180° − WCB → S 40° E

180°–270° (SW): RB = WCB − 180° → S 40° W

270°–360° (NW): RB = 360° − WCB → N 40° W

WCB: 0° to 360°, clockwise from north.
RB: 0° to 90°, from north or south, towards east or west.

The same four lines written both ways. The last rule (north-west quadrant) was asked in 2023 (Q141).

Fore and back bearing: the fore bearing (FB) is taken in the direction of progress, the back bearing (BB) in the opposite direction. In the whole circle system **BB = FB ± 180°**. In the reduced system the angle stays the same and both letters are reversed (N 32° E ↔ S 32° W).

Feature	Prismatic compass	Surveyor's compass
Needle	Broad needle, not used as an index	Edge-bar needle, acts as the index
Graduated ring	Attached to the needle; stays still while the box turns	Attached to the box; turns with the line of sight
Graduations	0° to 360°, **zero at the south end**, figures inverted	0° at N and S, 90° at E and W; **E and W interchanged**
Bearing read	Whole circle bearing	Reduced (quadrantal) bearing
Reading	Through the prism while sighting — sighting and reading together	By going round to the north end after sighting
Tripod	Not essential	Essential
Least count	30′	15′

Why zero is at the south end: the observer reads the ring through the prism at his own end, which is diametrically opposite the object. When the line of sight points north, the south end of the ring is under the prism — so it must carry the zero.

Declination, dip and local attraction

- **Magnetic declination** = angle between the true and magnetic meridians. **True bearing = magnetic bearing + east declination** (− west declination).
- **Isogonic lines** join places of equal declination; the **agonic line** joins places of zero declination.
- **Dip** = inclination of the needle to the horizontal: 0° at the magnetic equator, 90° at the magnetic poles. **Isoclinic lines** join places of equal dip; the **aclinic line** has zero dip. A sliding rider on the needle balances the dip.
- Variation of declination: diurnal (daily), annual, **secular** (the largest, over centuries) and irregular (magnetic storms).
- **Local attraction** — the needle is pulled aside by nearby iron, steel or electric cables. Test: if FB and BB of a line differ by exactly 180°, both its end stations are free. Start from such a line and correct the bearings at the affected stations.
- Check for a closed traverse: sum of interior angles = (2n − 4) × 90°. Included angles are not affected by local attraction, because both bearings at a station are deflected by the same amount.

4. Everyday analogy

A compass near a motorcycle or a steel almirah points the wrong way — that is local attraction. Every bearing read at that spot is wrong by the same amount, like a wall clock that runs 10 minutes fast: each reading is wrong, but the difference between two readings is still right.

5. Confusing areas and exam traps

- Prismatic: zero at **south**. Surveyor's: **E and W interchanged**. Do not mix the two.
- East declination is added, west declination subtracted, to get the true bearing from the magnetic bearing.
- Isogonic = equal declination; isoclinic = equal dip; agonic = zero declination; aclinic = zero dip.
- Local attraction changes bearings, not included angles.

6. Connections

Bearings feed directly into latitude and departure calculations (Topic 4). The true meridian is found by astronomical observation (Topic 10).

Memory code: "Prism reads South; Surveyor swaps East–West"

Prism reads South	Prismatic compass: zero at the south end, whole circle bearings
Surveyor swaps E–W	Surveyor's compass: E and W interchanged, reduced bearings
goNic – decliNation; cLinic – dip (incLination)	Isogonic and agonic → declination; isoclinic and aclinic → dip

Self-recall (close the notes, answer aloud, then check)

1. Convert WCB 250° to a reduced bearing.
   250° − 180° = 70° → S 70° W.
2. How do you detect local attraction?
   FB and BB of a line should differ by exactly 180°; if they do not, at least one of its stations is affected.
3. Magnetic bearing 100°, declination 3° E. True bearing?
   100° + 3° = 103°.

Quick check: 5 questions

1. The whole circle bearing of a line is 215°30′. Its reduced bearing is

   1. N 35°30′ E
   2. S 54°30′ W
   3. N 54°30′ W
   4. S 35°30′ W

   1. That is the opposite direction.
   2. Subtracts from 270° instead of subtracting 180°.
   3. Wrong quadrant.
   4. Correct. Third quadrant: RB = 215°30′ − 180° = 35°30′ → S 35°30′ W.

2. The fore bearing of a line AB is N 32° E. Its back bearing is

   1. S 32° W
   2. N 32° W
   3. S 58° W
   4. S 32° E

   1. Correct. In the reduced system the angle stays the same and both letters reverse.
   2. Only one letter is changed.
   3. The angle does not change.
   4. Only one letter is changed.

3. The magnetic bearing of a line is 48° and the magnetic declination is 2° W. The true bearing of the line is

   1. 50°
   2. 48°
   3. 46°
   4. 44°

   1. Adds a west declination — that is the rule for east declination.
   2. Ignores the declination.
   3. Correct. True bearing = magnetic bearing − west declination = 48° − 2° = 46°.
   4. Subtracts twice.

4. In a compass traverse the observed bearings are: AB — FB 60°, BB 240°; BC — FB 120°, BB 301°. Which station is affected by local attraction?

   1. A only
   2. C only
   3. B only
   4. A and B

   1. FB and BB of AB differ by exactly 180°, so A is free.
   2. Correct. A and B are free (240° − 60° = 180°). The FB of BC taken at B is therefore right, so its BB at C should be 300°; the 1° excess is local attraction at C.
   3. B is free, as line AB shows.
   4. Both are free.

5. Consider the following statements:

I. In a prismatic compass the zero of the graduated ring is at the south end.

II. A surveyor's compass gives reduced (quadrantal) bearings.

III. The dip of the needle is zero at the magnetic poles.

IV. Isogonic lines join places of equal magnetic declination.

Which of the above statements are correct?

1. I and II only
2. II and III only
3. I, II and IV only
4. I, II, III and IV

1. Leaves out IV, which is true.
2. III is false.
3. Correct. Dip is 90° at the magnetic poles and zero at the magnetic equator, so III is false.
4. Includes the false statement III.

Topic 4: Theodolite, Traversing and Computation of Coordinates

Syllabus line: *Theodolites — types; establishment of control by traversing; computation of coordinates.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): no direct question in this shift

- This paper (22 Aug 2023, Shift 2) had no question on the theodolite or traverse computations. They are named in the syllabus and are standard in other AEE papers, so learn them fully.

1. Basic idea

A **theodolite** measures horizontal and vertical angles precisely. A **traverse** is a chain of connected lines whose lengths and directions are measured. From these we compute the **coordinates** of every station.

2. Why it is needed

A compass reads only to 30′. Control points for a town, a canal or a bridge need angles correct to seconds. Coordinates let us plot each station independently, so a plotting error at one station is not carried forward.

3. How it works

Type	Feature
Transit theodolite	Telescope can be turned through a full circle in the vertical plane (transited)
Non-transit theodolite	Telescope cannot be transited — obsolete
Vernier theodolite	Circles read by verniers, usually to 20″
Optical (glass arc) theodolite	Read by an optical micrometer to 1″
Electronic (digital) theodolite	Angles shown on a display; forms part of a total station

Fundamental lines and their required relations

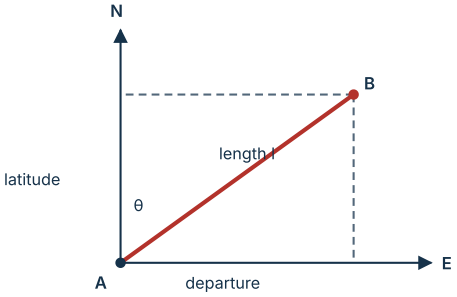
- Axis of the plate level ⊥ vertical axis.
- Line of collimation ⊥ horizontal (trunnion) axis.
- Horizontal axis ⊥ vertical axis.
- Axis of the altitude level ∥ line of collimation.
- **Temporary adjustments** at every set-up: centring over the station → levelling → focusing the eyepiece → focusing the objective (removes parallax).

Measuring a horizontal angle: the **repetition method** adds the same angle several times on the circle and divides — used for one angle of high precision. The **reiteration (direction) method** measures several angles round a station in turn and closes the horizon.

Error	Removed by
Line of collimation not ⊥ horizontal axis	Mean of face-left and face-right readings

Error	Removed by
Horizontal axis not ⊥ vertical axis	Mean of face-left and face-right readings
Index error of the vertical circle	Mean of face-left and face-right readings
Eccentricity of the circles or verniers	Reading both verniers
Graduation errors	Repetition, or using different parts of the circle
Vertical axis not truly vertical (plate level out of adjustment)	**Not removed** by changing face — level carefully

Traverse: a **closed** traverse returns to its starting point or ends on another known point; an **open** traverse does not (roads, canals) and has no check. Angles may be measured as included angles, deflection angles or direct bearings.



Latitude L = l cos θ
north positive (northing), south negative

Departure D = l sin θ
east positive (easting), west negative

Closed traverse: ΣL = 0 and ΣD = 0
Closing error $e = \sqrt{[(\Sigma L)^2 + (\Sigma D)^2]}$
Direction: $\tan \delta = \Sigma D / \Sigma L$
Relative accuracy = e / perimeter

θ is the reduced bearing of the line. Latitude is the projection on the north–south line; departure is the projection on the east–west line.

Checks and balancing

- Sum of interior angles = (2n − 4) × 90°; sum of exterior angles = (2n + 4) × 90°; algebraic sum of deflection angles = 360°.
- **Bowditch (compass) rule:** correction to the latitude (or departure) of a line = total error in latitude (or departure) × length of that line ÷ perimeter, with opposite sign. Assumes angles and lengths are measured with equal precision. Most commonly used.
- **Transit rule:** correction to latitude = total error × latitude of that line ÷ arithmetic sum of all latitudes. Used when angles are more precise than lengths.
- **Consecutive coordinates** = latitude and departure of each line. **Independent coordinates** = total latitude and total departure of each station from a common origin: $N_B = N_A + L_{AB}$, $E_B = E_A + D_{AB}$.

4. Everyday analogy

Walk round a field and come back to the tree you started from: the total distance you moved north must equal the total you moved south, and the same for east and west. If your notes do not balance, the small gap is the closing error.

5. Confusing areas and exam traps

- Latitude uses **cos**, departure uses **sin** — when θ is measured from the north–south line.
- Bowditch → proportional to **length**; Transit → proportional to the **latitude or departure** itself.
- Changing face does **not** remove the error of a vertical axis that is not truly vertical.
- Repetition suits one angle; reiteration suits many angles at one station.

6. Connections

Coordinates are the output of a total station (Topic 7) and of GPS (Topic 10). The closing-error idea is used again in levelling (Topic 5) and in the adjustment of triangulation (Topic 8).

Memory code: "LC – DS"

LC	Latitude = l Cos θ
DS	Departure = l Sin θ
B-Length, T-Lat	Bowditch in proportion to Length; Transit in proportion to the Latitude or departure itself

Self-recall (close the notes, answer aloud, then check)

1. What are the temporary adjustments of a theodolite, in order?
   Centring, levelling, focusing the eyepiece, focusing the objective (eliminating parallax).

2. State the conditions for a closed traverse to close.
   $\Sigma L = 0$ and $\Sigma D = 0$ (and the angles must satisfy $(2n - 4) \times 90°$).
3. Which error is not removed by face-left and face-right observations?
   The error due to the vertical axis not being truly vertical.

Quick check: 5 questions

1. A traverse line is 200 m long and its bearing is N 30° E. Its latitude and departure are

   1. 100.0 m N and 173.2 m E
   2. 173.2 m S and 100.0 m W
   3. 173.2 m N and 100.0 m E
   4. 141.4 m N and 141.4 m E

   1. Swaps cos and sin.
   2. Right magnitudes, wrong signs — the line runs north and east.
   3. Correct. L = 200 cos 30° = 173.2 m (north); D = 200 sin 30° = 100 m (east).
   4. That would be a bearing of N 45° E.

2. In a closed traverse of perimeter 1000 m, ΣL = +0.30 m and ΣD = −0.40 m. The closing error and the relative accuracy are

   1. 0.70 m and 1 in 1429
   2. 0.50 m and 1 in 2000
   3. 0.10 m and 1 in 10,000
   4. 0.50 m and 1 in 1000

   1. Adds the two errors instead of taking the root of the sum of squares.
   2. Correct. $e = \sqrt{(0.3^2 + 0.4^2)} = 0.50$ m; relative accuracy = 0.5/1000 = 1 in 2000.
   3. Subtracts the two errors.
   4. The error is right but 0.5/1000 = 1/2000.

3. The sum of the interior angles of a closed traverse of six sides should be

   1. 540°
   2. 900°
   3. 1080°
   4. 720°

   1. That is for five sides.
   2. That is for seven sides.
   3. That is the sum of exterior angles of a hexagon: $(2n + 4) \times 90°$.
   4. Correct. $(2n - 4) \times 90° = 8 \times 90° = 720°$.

4. Consider the following statements about a transit theodolite:
   I. Taking the mean of face-left and face-right readings removes the error due to the line of collimation not being perpendicular to the horizontal axis.
   II. It also removes the error due to the horizontal axis not being perpendicular to the vertical axis.
   III. It also removes the error due to the vertical axis not being truly vertical.
   IV. Reading both verniers removes the error due to eccentricity.

   Which of the above statements are correct?

   1. I, II and IV only
   2. I and II only
   3. II and III only
   4. I, II, III and IV

   1. Correct. I, II and IV are true. The vertical-axis error (III) is not removed by changing face.
   2. Leaves out IV, which is true.
   3. III is false.
   4. Includes the false statement III.

5. **Assertion (A):** Bowditch's rule distributes the closing error in proportion to the lengths of the traverse sides.
Reason (R): Bowditch's rule assumes that angular measurements are more precise than linear measurements.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. R is false.
2. R is false.
3. Correct. A is true. Bowditch assumes angles and lengths are equally precise; the transit rule is the one for more precise angles.
4. A is true.

Topic 5: Measurement of Elevations — Levelling

Syllabus line: *Measurement of elevations, spirit and trigonometric levelling.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 1 question

- Match the description with the type of levelling: pressure difference → barometric; vertical angles and distances → trigonometric; horizontal line of sight → spirit; points at short intervals along a fixed line → profile (Q132).

1. Basic idea

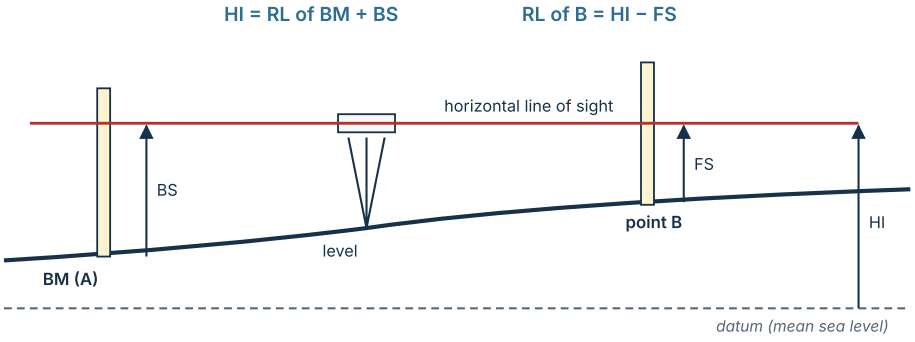
Levelling finds how much higher or lower one point is than another. A **level** gives a horizontal line of sight; a graduated **staff** held on each point is read against that line. The difference between two staff readings is the difference in level.

2. Why it is needed

Water flows only downhill. Canals, sewers, roads and railway gradients, building plinths and reservoir capacities all depend on accurate levels.

3. How it works

Term	Meaning
Level surface	Curved surface everywhere perpendicular to the direction of gravity, such as still water
Datum	Reference surface for heights — in India, mean sea level
Reduced level (RL)	Height of a point above (or below) the datum
Bench mark (BM)	Fixed point of known RL — GTS, permanent, arbitrary or temporary
Back sight (BS)	First reading after setting up, taken on a point of known RL
Fore sight (FS)	Last reading before shifting the instrument
Intermediate sight (IS)	Any reading between the BS and the FS
Change (turning) point	Point on which both an FS and a BS are taken
Height of instrument (HI)	RL of the line of sight



The level gives one horizontal line. The staff reading is larger on lower ground, so a bigger reading means a lower point.

Method of reduction	Rule	Arithmetic check	Use
Height of instrument (collimation)	HI = RL + BS; RL = HI − IS or FS	ΣBS − ΣFS = last RL − first RL	Quick; good when there are many intermediate sights (profiles). Intermediate RLs are not checked
Rise and fall	Previous reading − next reading: positive = rise, negative = fall	ΣBS − ΣFS = ΣRise − ΣFall = last RL − first RL	Slower, but checks every RL including intermediate sights

Curvature and refraction

- Curvature correction $C_c = d^2/2R =$ **$0.0785\ d^2$** m (d in km) — subtracted from the staff reading.
- Refraction correction C_r = 1/7 of the curvature correction = $0.0112\ d^2$ m — added.
- Combined correction = **$0.0673\ d^2$** m — subtracted. Objects appear lower than they really are.
- Distance to the visible horizon from a height h metres: **$d = 3.855\sqrt{h}$** km.
- Keeping the back-sight and fore-sight distances equal cancels curvature, refraction and the collimation error of the level.

Type of levelling	What it does
Simple levelling	Two points visible from one set-up
Differential (fly) levelling	Points far apart; several set-ups with change points
Profile (longitudinal) levelling	Levels at short intervals along a fixed line such as a road or canal centre line
Cross-sectioning	Levels at right angles to the centre line
Reciprocal levelling	Across a river or valley; readings from both banks; the mean removes collimation, curvature and refraction errors
Check levelling	At the end of the day's work, back to the starting point
Trigonometric levelling	Elevations computed from vertical angles and measured distances
Barometric levelling	From the difference of atmospheric pressure at two points

Sensitivity of the level tube

- Angular value of one division: $\alpha = s / (n\,D)$ radians (× 206265 for seconds), where s = change in staff reading when the bubble moves n divisions and D = distance to the staff.
- Sensitivity increases with a larger radius of the tube, a larger diameter, a longer bubble, and a liquid of lower viscosity and surface tension.
- An inverted staff reading (staff held upside down against a ceiling or beam) is entered as negative.

4. Everyday analogy

A mason checks a wall with a transparent tube filled with water: the water stands at the same height at both ends. A level does the same job with a line of sight, and the staff is the measuring scale held at each end.

5. Confusing areas and exam traps

- A **larger** staff reading means a **lower** point. Students often reverse rise and fall.
- Curvature makes objects look lower (correction negative); refraction makes them look higher (correction positive). The net effect is negative.
- The HI method does not check intermediate sights; the rise-and-fall method does.
- A larger radius of the level tube means higher sensitivity.

6. Connections

Levels are the raw data for contours (Topic 6). The $0.0673\ d^2$ rule also fixes the heights of triangulation signals (Topic 8). Trigonometric levelling uses the theodolite (Topic 4).

Memory code: “Back adds, Fore subtracts”

Back adds	HI = RL + Back sight

Fore subtracts	RL = HI − Fore sight (or intermediate sight)
785 − 112 = 673	0.0785 d^2 (curvature) − 0.0112 d^2 (refraction) = 0.0673 d^2 (combined)

Self-recall (close the notes, answer aloud, then check)

1. Write the arithmetic check for the rise-and-fall method.
   ΣBS − ΣFS = ΣRise − ΣFall = last RL − first RL.
2. What is the combined correction for curvature and refraction for a 1 km sight?
   0.0673 m, subtracted from the staff reading.
3. Why is reciprocal levelling used across a river?
   The level cannot be set midway, so readings are taken from both banks; the mean cancels collimation error, curvature and refraction.

Quick check: 5 questions

1. The RL of a bench mark is 100.000 m. The back sight on it is 1.500 m and the fore sight on point B is 2.250 m. The RL of B is

   1. 100.750 m
   2. 103.750 m
   3. 96.250 m
   4. 99.250 m

   1. Adds the fall instead of subtracting it.
   2. Adds both readings.
   3. Subtracts both readings.
   4. Correct. HI = 100.000 + 1.500 = 101.500; RL of B = 101.500 − 2.250 = 99.250 m.

2. The combined correction for curvature and refraction for a sight of 2 km is about

   1. 0.269 m
   2. 0.135 m
   3. 0.314 m
   4. 0.067 m

   1. Correct. 0.0673 × 2^2 = 0.269 m.
   2. Multiplies by d instead of d^2.
   3. That is the curvature correction alone (0.0785 × 4).
   4. That is the value for 1 km.

3. In reciprocal levelling, with the level near A the staff readings were A = 1.500 m and B = 2.500 m; with the level near B they were A = 0.800 m and B = 1.600 m. If the RL of A is 100.000 m, the RL of B is

   1. 99.000 m
   2. 99.200 m
   3. 99.100 m
   4. 100.900 m

   1. Uses only the first set of readings.
   2. Uses only the second set.
   3. Correct. Apparent differences are 1.000 m and 0.800 m; true difference = mean = 0.900 m. B has the larger readings, so B is lower: 99.100 m.
   4. B is lower than A, not higher.

4. Match List-I (type of levelling) with List-II (use):

List-I	List-II
A. Reciprocal levelling	1. Along the centre line of a road or canal
B. Profile levelling	2. Verifying the day's work
C. Barometric levelling	3. Across a wide river
D. Check levelling	4. From atmospheric pressure

1. A-3, B-1, C-4, D-2
2. A-1, B-3, C-4, D-2
3. A-3, B-1, C-2, D-4
4. A-4, B-1, C-3, D-2

1. Correct. Reciprocal → across a wide river; profile → along a centre line; barometric → from air pressure; check → verifying the day's work.
2. Swaps reciprocal and profile levelling.
3. Swaps barometric and check levelling.
4. Reciprocal levelling has nothing to do with pressure.

5. Consider the following statements:
I. The rise-and-fall method provides a check on intermediate sights.
II. The height-of-instrument method is quicker when there are many intermediate sights.
III. Keeping back-sight and fore-sight distances equal eliminates the collimation error.
IV. The sensitivity of a level tube decreases as its radius increases.

Which of the above statements are correct?

1. I and II only
2. I, II and III only
3. II, III and IV only
4. I, II, III and IV

1. Leaves out III, which is true.
2. Correct. I, II and III are true. Sensitivity increases with the radius of the tube, so IV is false.
3. IV is false.
4. Includes the false statement IV.

Topic 6: Contours and Digital Elevation Modelling

Syllabus line: *Contours, digital elevation modelling concept.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): no direct question in this shift

- Not asked in this paper. Contour characteristics are a standard one-mark area in AEE and AE papers, and 'digital elevation modelling' is named in the syllabus.

1. Basic idea

A **contour** is an imaginary line joining points of equal elevation. Imagine a hill standing in a lake: each time the water level rises by one metre, the new shoreline is a contour. Drawn on a map, contours show the shape of the ground in three dimensions on a flat sheet.

2. Why it is needed

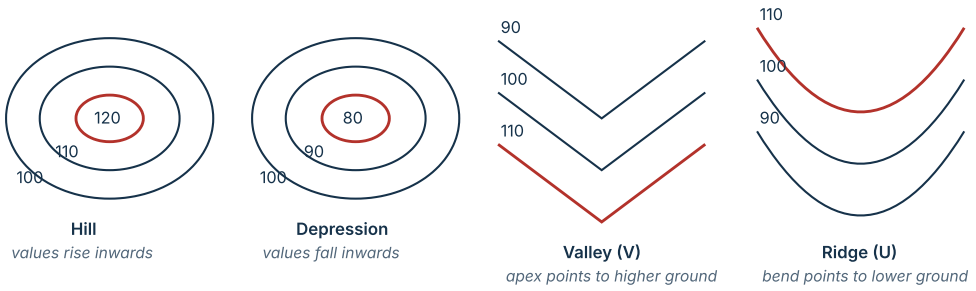
From a contour map an engineer can pick a road or canal alignment at a chosen gradient, find a reservoir's capacity, mark a catchment area, compute earthwork and check whether two points are intervisible — all without going back to the field.

3. How it works

Two key terms

- **Contour interval** — the vertical distance between two consecutive contours. It is kept **constant** on a map.
- **Horizontal equivalent** — the horizontal distance between two consecutive contours. It **varies** with the slope.
- Gradient between two contours = contour interval ÷ horizontal equivalent.

- The contour interval is chosen smaller for a larger map scale, flatter ground, more important work, and when more time and money are available.



Read the numbers, not the shape alone: a hill and a depression look the same until you see which way the values increase.

Characteristics of contours

- Every contour closes on itself, inside or outside the map.
- Contours of different elevations never cross, except at an **overhanging cliff** or cave; they merge into one line at a **vertical cliff**.
- Close spacing = steep slope; wide spacing = gentle slope; equal spacing = uniform slope; straight, parallel and equally spaced = a plane surface.
- Contours cross ridge lines and valley lines at **right angles**.
- A **saddle** (pass) lies between two hills, with higher ground on two opposite sides and lower ground on the other two.

Method of contouring	How	Remark
Direct method	Points on each contour are actually traced and marked on the ground with a level, then surveyed	Most accurate; slow and costly; for small areas
Indirect — squares (grid)	Spot levels at the corners of a grid; contours interpolated	Small, fairly level areas
Indirect — cross-sections	Levels along lines at right angles to a centre line	Route surveys: roads, canals, railways
Indirect — tacheometric (radial lines)	Distances and levels read by tacheometer along radial lines	Hilly ground

Interpolation of contours between spot levels is done by estimation, by arithmetic calculation (most accurate) or graphically, assuming a uniform slope between the two points.

Digital elevation modelling — the concept

- A **digital elevation model (DEM)** stores ground elevation as numbers a computer can process, instead of as lines drawn on paper.
- **Raster (grid) DEM** — a matrix of square cells, each holding one elevation. Simple; the cell size fixes the detail.
- **TIN (triangulated irregular network)** — irregularly spaced points joined into non-overlapping triangles; more points where the ground is rough.
- **DSM** (digital surface model) includes the tops of buildings and trees; **DTM** (digital terrain model) is the bare earth.
- Sources: ground survey (total station, GPS), photogrammetry, LiDAR, radar interferometry (SRTM), satellite stereo images (CartoDEM from Cartosat).
- Products: slope and aspect maps, contours, watershed and drainage delineation, cut-and-fill volumes, visibility analysis, flood mapping.

4. Everyday analogy

Slice a boiled potato into layers of equal thickness and trace each slice on paper, one inside the other: you have drawn its contour map. Thin, crowded outlines mark the steep side of the potato.

5. Confusing areas and exam traps

- **Valley:** V-shaped, apex towards higher ground. **Ridge:** U-shaped, bend towards lower ground.
- Contour interval is constant on one map; horizontal equivalent is not.
- Contours merge at a vertical cliff; they cross only at an overhang.

- DSM includes buildings and trees; DTM does not.

6. Connections

Contour gradient → highway and canal alignment (Transport and Irrigation modules). Area enclosed by contours → reservoir capacity (Hydrology). DEMs come from photogrammetry and remote sensing (Topics 9 and 10).

Memory code: "Valley points Up, Ridge points Down"

Valley points Up	V-shaped contours with the apex towards higher ground (upstream)
Ridge points Down	U-shaped contours with the bend towards lower ground
Cliff merges, Overhang crosses	Vertical cliff → contours join; overhanging cliff → contours cross

Self-recall (close the notes, answer aloud, then check)

1. Define contour interval and horizontal equivalent.
   Contour interval = vertical distance between consecutive contours (constant). Horizontal equivalent = horizontal distance between them (varies with slope).
2. How do you tell a hill from a depression on a contour map?
   Closed contours with values increasing inwards show a hill; values decreasing inwards show a depression.
3. What is a TIN?
   A terrain model made of non-overlapping triangles formed from irregularly spaced elevation points.

Quick check: 5 questions

1. Consider the following statements about contours:
   I. Contours of different elevations cross each other only at an overhanging cliff.
   II. Closely spaced contours indicate a steep slope.
   III. In a valley the contours are V-shaped with the apex pointing towards lower ground.
   IV. Every contour closes on itself somewhere.

   Which of the above statements are correct?

   1. I and II only
   2. I, II and IV only
   3. II, III and IV only
   4. I, II, III and IV

   1. Leaves out IV, which is true.
   2. Correct. I, II and IV are true. In a valley the apex points towards higher ground, so III is false.
   3. III is false.
   4. Includes the false statement III.

2. On a map of scale 1 : 5000 with a contour interval of 5 m, two consecutive contours are 2 cm apart. The gradient of the ground between them is

   1. 1 in 10
   2. 1 in 50
   3. 1 in 20
   4. 1 in 100

   1. Would need 1 cm between the contours.
   2. Would need 5 cm between the contours.
   3. Correct. Horizontal equivalent = 2 cm × 5000 = 100 m; gradient = 5/100 = 1 in 20.
   4. Would need 10 cm between the contours.

3. A series of closed contours in which the values increase towards the centre represents

   1. a depression
   2. a saddle
   3. a ridge line
   4. a hill

   1. A depression has values decreasing towards the centre.
   2. A saddle is not a set of closed loops.
   3. A ridge shows U-shaped, not closed, contours.
   4. Correct. Higher values inside closed contours mean a hill.

4. S1: A TIN represents terrain by non-overlapping triangles formed from irregularly spaced points.
S2: A digital surface model (DSM) gives bare-earth elevations after buildings and trees are removed.

1. Both S1 and S2 are true
2. S1 is true and S2 is false
3. S1 is false and S2 is true
4. Both S1 and S2 are false

1. S2 is false.
2. Correct. S1 is true. Bare earth is the DTM; a DSM includes the tops of buildings and trees.
3. S1 is true.
4. S1 is true.

5. **Assertion (A):** The direct method of contouring is the most accurate but is slow.
Reason (R): In the direct method the points lying on each contour are actually located on the ground before being surveyed.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. Correct. Both are true, and tracing each contour on the ground is what makes the method accurate and slow.
2. R is the reason for A.
3. R is true.
4. A is true.

Topic 7: Indirect Distance Measurement — Tacheometry, EDM and Total Station

Syllabus line: *Measurement of distances — indirect methods, optical and electronic devices.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 1 question

- Statement–conclusion on stadia tacheometry: the paper **swapped** the descriptions of the fixed-hair and movable-hair methods, so neither conclusion was correct (Q133).

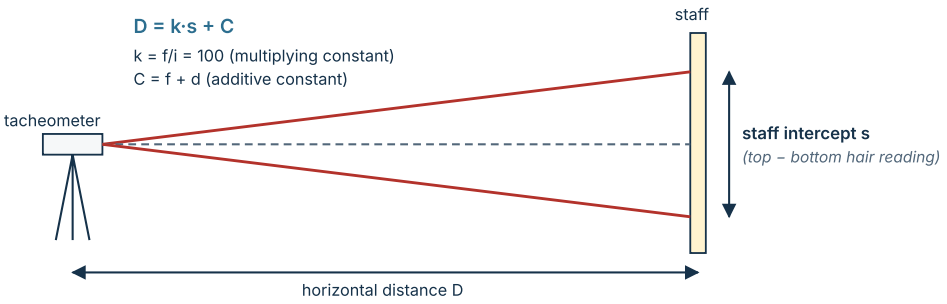
1. Basic idea

Tacheometry finds horizontal distances and differences of level from instrument readings alone, without a chain. The telescope has two extra horizontal hairs (**stadia hairs**); the length of staff seen between them, the **staff intercept**, grows in proportion to the distance.

2. Why it is needed

On broken, hilly or marshy ground, and across rivers and ravines, chaining is slow or impossible. Tacheometry is fast and good enough for contouring and preliminary route surveys (accuracy about 1 in 1000). Electronic distance measurement has now taken over for precise work.

3. How it works



The two stadia rays open out at a fixed angle, so the length of staff they cut is proportional to the distance — twice as far, twice the intercept.

System	What is fixed	What varies
Fixed-hair stadia (the common method)	Distance between the stadia hairs (stadia interval)	Staff intercept — it changes with the distance

System	What is fixed	What varies
Movable-hair (subtense) stadia	Staff intercept — set by two targets a fixed distance apart on the staff	Stadia interval — the hairs are moved by a micrometer screw
Tangential method	No stadia hairs used	Two vertical angles are read to two targets on the staff
Subtense bar	Horizontal bar of fixed length, usually 2 m	The horizontal angle it subtends is measured with a theodolite

Tacheometric formulas

- Horizontal line of sight: **$D = k\,s + C$**. With an **anallactic lens** the additive constant C becomes zero, so $D = 100\,s$.
- Inclined sight, staff vertical: **$D = k\,s\cos^2\theta + C\cos\theta$** and vertical component **$V = (k\,s\sin 2\theta)/2 + C\sin\theta$**.
- RL of staff station = RL of instrument axis ± V − central hair reading (+ for an angle of elevation, − for depression).
- Subtense bar: $D = (b/2)\cot(\alpha/2) \approx b/\alpha$ (α in radians, b = length of bar).

Electronic distance measurement (EDM)

- **Principle:** the instrument sends a modulated electromagnetic wave to a reflector and measures the **phase difference** between the outgoing and returning wave. Distance = whole number of half wavelengths + the fraction from the phase difference.
- **Microwave** instruments (Tellurometer) — long range, up to about 100 km; need an instrument at both ends.
- **Visible-light** instruments (Geodimeter) — medium range, about 25 km at night.
- **Infrared** instruments (Distomat) — short range, about 2–5 km; the kind built into total stations.
- Corrections: atmospheric (temperature and pressure change the wave velocity), prism constant, slope and height.

Total station. It measures three things — horizontal angle, vertical angle and slope distance — and computes the rest: horizontal distance, difference of level, and the coordinates (northing, easting, elevation) of each point. Data are stored digitally and downloaded to a computer. It is used for traversing, topographic detail, setting out, and remote height and missing-line measurement. Reflectorless models measure to a wall or rock face without a prism.

4. Everyday analogy

Hold your thumb at arm's length against a distant building. The farther the building, the more floors your thumb covers. The stadia hairs are a fixed "thumb" inside the telescope, and the staff is the building with floors marked on it.

5. Confusing areas and exam traps

- **Fixed hair:** interval fixed, intercept varies. **Movable hair:** intercept fixed, interval varies. The 2023 paper tested this by swapping them.
- The anallactic lens makes the additive constant zero; it does not change the multiplying constant.
- For an inclined sight D uses $\cos^2\theta$, V uses $\sin 2\theta / 2$.
- EDM measures the **slope** distance; the horizontal distance is computed.

6. Connections

Tacheometric radial lines are one indirect method of contouring (Topic 6). EDM made trilateration practical (Topic 8). Total-station coordinates use the latitude–departure idea of Topic 4.

Memory code: “Fixed hair, Free staff”

Fixed hair, Free staff	Fixed-hair method: hairs fixed, staff intercept free to vary
Moving hair, Marked staff	Movable-hair method: hairs move, staff intercept fixed by targets
A for A	Anallactic lens → Additive constant zero

Self-recall (close the notes, answer aloud, then check)

1. Write the tacheometric formula for a horizontal sight.
   $D = k\,s + C$; $k = f/i$ (usually 100), $C = f + d$ (zero with an anallactic lens).
2. What does an EDM actually measure?

The phase difference between the transmitted and the reflected modulated wave, from which the slope distance is computed.

3. Which three quantities does a total station measure directly?
Horizontal angle, vertical angle and slope distance.

Quick check: 5 questions

1. A tacheometer fitted with an anallactic lens (k = 100) is sighted horizontally on a vertical staff. The stadia hair readings are 1.200 m and 2.100 m. The horizontal distance is

1. 45 m
2. 90 m
3. 165 m
4. 210 m

1. Uses half the intercept.
2. Correct. s = 2.100 − 1.200 = 0.900 m; D = 100 × 0.9 = 90 m.
3. Uses the mean of the readings instead of their difference.
4. Uses the upper reading alone.

2. With k = 100 and C = 0, the staff intercept on a vertical staff is 2.0 m when the line of sight is inclined at 30° to the horizontal. The horizontal distance is

1. 150 m
2. 173.2 m
3. 100 m
4. 200 m

1. Correct. $D = k\,s\cos^2\theta = 100 \times 2 \times (0.866)^2 = 150$ m.
2. Uses $\cos\theta$ instead of $\cos^2\theta$.
3. Uses $\sin\theta$.
4. Ignores the inclination.

3. **Statement:** Two forms of the stadia method are in use — fixed hair and movable hair.
Conclusion 1: In the fixed-hair method the stadia interval is constant and the staff intercept varies with the distance.
Conclusion 2: In the movable-hair method the staff intercept is kept constant and the stadia interval is varied.

1. Only conclusion 1 is correct
2. Only conclusion 2 is correct
3. Both conclusions 1 and 2 are correct
4. Neither conclusion 1 nor 2 is correct

1. Conclusion 2 is also correct.
2. Conclusion 1 is also correct.
3. Correct. Both descriptions are right. (In 2023 Q133 the two were swapped, so neither was correct there.)
4. Both are correct as written here.

4. An anallactic lens is provided in a tacheometer to

1. increase the magnification of the telescope
2. make the multiplying constant equal to 1
3. remove parallax
4. make the additive constant zero

1. Magnification depends on the objective and eyepiece.
2. The multiplying constant is kept at 100.
3. Parallax is removed by focusing.
4. Correct. It brings the vertex of the stadia rays to the instrument axis, so C = 0 and D = k s.

5. Consider the following statements:

I. An EDM finds distance from the phase difference of a modulated wave.

II. A total station combines an electronic theodolite, an EDM and a microprocessor.

III. The Tellurometer uses microwaves.

IV. The Geodimeter uses sound waves.

Which of the above statements are correct?

1. I and II only
2. I, II and III only
3. II, III and IV only
4. I, II, III and IV

1. Leaves out III, which is true.
2. Correct. I, II and III are true. The Geodimeter uses visible light, so IV is false.
3. IV is false.
4. Includes the false statement IV.

Topic 8: Triangulation and Adjustment of Observations

Syllabus line: *Establishment of control by triangulation; measurement and adjustment of observations.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): no direct question in this shift

- Not asked in this paper. The syllabus names both triangulation and adjustment of observations, so the classification, the figures and the rules for most probable values must be known.

1. Basic idea

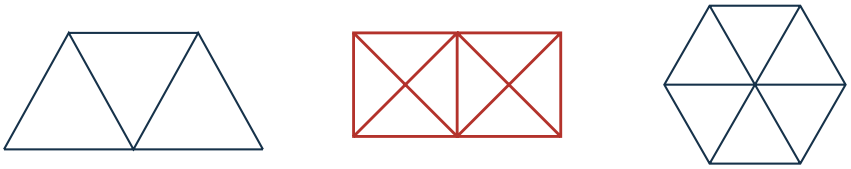
In **triangulation** the area is covered by a network of triangles. Only **one side** (the base line) is measured very accurately; all the **angles** are measured; every other side is then computed by trigonometry (sine rule). The corners become control points for all later surveys.

2. Why it is needed

Before EDM, measuring a long line precisely was far harder than measuring an angle. Triangulation carried accurate control across a whole country from a few measured base lines — the Great Trigonometrical Survey of India was done this way. Because no measurement is perfect, the observations must then be **adjusted** so that they agree with geometry.

3. How it works

Order	Average triangle closure	Length of base	Length of sides	Purpose
Primary (first order)	Less than 1″	5–15 km	30–150 km	Shape of the earth; control for the whole country
Secondary (second order)	About 3″	1.5–5 km	8–65 km	Fills in the primary network
Tertiary (third order)	About 6″	0.5–3 km	1.5–10 km	Control for detailed topographic surveys



Single chain of triangles
fast, narrow strip, fewest checks

Braced quadrilaterals
strongest, most checks

Centred polygon
covers a wide area quickly

Each station is a triangle corner. More lines crossing the same figure give more geometric conditions, and so more checks on the angles.

Field terms: a **well-conditioned triangle** has no angle below 30° or above 120° (the best shape is isosceles with base angles of about 56°). **Signals** mark stations for sighting — opaque (poles, targets) or luminous (the **heliotrope** reflects

sunlight; lamps at night). A **satellite (eccentric) station** is used when the instrument cannot be set over the true station, such as a church spire; the readings are then **reduced to centre**. **Trilateration** measures the sides with EDM instead of the angles.

Laws of accidental errors (normal distribution)

- Small errors occur more often than large ones.
- Positive and negative errors of the same size are equally likely.
- Very large errors do not occur.

Most probable value and weights

- The **most probable value** (MPV) of a quantity measured several times with equal care is the **arithmetic mean**.
- With unequal reliability each observation gets a **weight**; MPV = $\Sigma(w\,x) / \Sigma w$ (the weighted mean). Weight is inversely proportional to the variance (square of the probable error).
- **Principle of least squares:** the MPV is the value for which the sum of the squares of the residuals (weighted) is a minimum.
- Probable error of a single observation = $0.6745\,\sqrt{[\Sigma v^2 / (n-1)]}$; of the mean = $0.6745\,\sqrt{[\Sigma v^2 / n(n-1)]}$.
- Weight of the mean of n observations of unit weight = n. If a quantity of weight w is multiplied by k, the weight becomes w/k^2. For a sum or difference, $1/w = 1/w_1 + 1/w_2$.

Adjusting angles

- **Station adjustment:** the angles closing the horizon at a station must add up to 360°.
- **Figure adjustment:** the three angles of a plane triangle must add up to 180° (plus the spherical excess for large triangles, about 1″ per 196 km²).
- Equal weights → distribute the error **equally**. Unequal weights → distribute the error in **inverse proportion to the weights** (the less reliable angle takes the larger correction).

4. Everyday analogy

Three friends measure the same table as 150.2, 150.4 and 150.6 cm. The fairest single answer is the mean, 150.4 cm. If one friend used a steel rule and is twice as reliable, his reading is counted twice — that is a weight of 2.

5. Confusing areas and exam traps

- Triangulation: angles measured, one side measured. **Trilateration: sides measured.**
- Braced quadrilaterals are the strongest figure; a single chain of triangles is the weakest.
- Corrections go in **inverse** proportion to the weights, not in direct proportion.
- Residual = observed value − most probable value; true error = observed value − true value (never exactly known).

6. Connections

Well-conditioned triangles were first met in chain surveying (Topic 2). Signal heights use the $0.0673\,d^2$ rule of Topic 5. GPS has replaced triangulation for new control (Topic 10), but the adjustment rules stay the same.

Memory code: “One Base, All Angles”

One Base, All Angles	Triangulation: measure one base line and every angle; compute the other sides
1 – 3 – 6	Average triangle closure: primary 1″, secondary 3″, tertiary 6″
Weak takes more	The observation with the smaller weight receives the larger correction

Self-recall (close the notes, answer aloud, then check)

1. What is measured in triangulation, and what in trilateration?
   Triangulation: all angles and one base line. Trilateration: the sides (by EDM).
2. State the principle of least squares.
   The most probable value is the one that makes the sum of the squares of the (weighted) residuals a minimum.
3. How is the angular error of a triangle distributed when the angles have unequal weights?
   In inverse proportion to the weights.

Quick check: 5 questions

1. For a triangle to be well-conditioned, its angles should lie between
   1. 10° and 170°
   2. 45° and 90°
   3. 60° and 150°
   4. 30° and 120°

1. Angles as small as 10° make the computed sides very sensitive to small errors.
2. This range is narrower than needed.
3. 150° is too large.
4. Correct. No angle should be less than 30° or more than 120°.

2. An angle was measured as 30°10′20″ with weight 2 and as 30°10′26″ with weight 1. Its most probable value is
   1. 30°10′21″
   2. 30°10′23″
   3. 30°10′22″
   4. 30°10′24″

1. Too close to the first reading.
2. That is the plain mean; it ignores the weights.
3. Correct. Weighted mean of the seconds = (2 × 20 + 1 × 26)/3 = 22″.
4. Gives the larger weight to the wrong reading.

3. The three angles of a plane triangle, measured with equal weight, add up to 180°00′12″. The correction to be applied is
   1. −4″ to each angle
   2. +4″ to each angle
   3. −12″ to the largest angle
   4. −6″ to the two smaller angles

1. Correct. The sum is 12″ too large; with equal weights the correction is shared equally: −4″ each.
2. The sum is too large, so the correction is negative.
3. Equal weights mean an equal share.
4. Equal weights mean an equal share among all three.

4. Consider the following statements:
   I. Braced quadrilaterals give the strongest triangulation figures.
   II. A satellite station is used when the instrument cannot be set up over the main station.
   III. A heliotrope is a luminous signal that reflects sunlight.
   IV. In trilateration all the angles are measured and only one side.

   Which of the above statements are correct?
   1. I and II only
   2. I, II and III only
   3. II, III and IV only
   4. I, II, III and IV

1. Leaves out III, which is true.
2. Correct. I, II and III are true. Statement IV describes triangulation; in trilateration the sides are measured.
3. IV is false.
4. Includes the false statement IV.

5. An angle is measured nine times, each observation having unit weight. The weight of the arithmetic mean is
   1. 3
   2. 1/9
   3. 81
   4. 9

1. That is √n; the probable error falls by this factor.
2. Weight rises, not falls, with more observations.
3. That is n^2.
4. Correct. The weight of the mean of n observations of unit weight is n = 9.

Topic 9: Plane Tabling, Photogrammetry and Map Substitutes

Syllabus line: *Map preparation by plane tabling and by photogrammetry; map substitutes.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 2 questions

- Scale of an aerial photograph: f = 200 mm, flying height 2700 m, ground elevation 200 m → 1 in 12,500 (Q137).

- Flight planning, statement–conclusion: 200 mm × 200 mm photo at 1 : 10,000 covers 4 km² (not 9.6 km²), and 192 km² needs 200 photos (not 20) → neither conclusion correct (Q139).

1. Basic idea

Both methods produce a map without first writing long field books. On a **plane table** the map is drawn in the field while looking at the ground. In **photogrammetry** the ground is photographed (usually from the air) and the measurements are made on the photographs.

2. Why they are needed

Plane tabling is quick and cheap for small-scale detail work, and nothing is forgotten because the surveyor sees the ground while plotting. Photogrammetry maps large or inaccessible areas — forests, hills, flood plains — in a fraction of the time of ground survey.

3. How it works

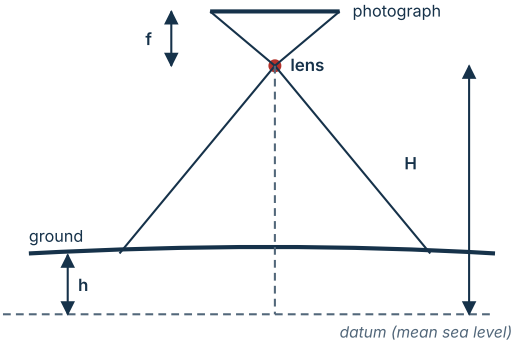
Plane tabling

Accessories: drawing board on a tripod, **alidade** (plain or telescopic) for sighting and ruling, spirit level, **trough compass** for marking north, and **plumbing fork (U-fork)** with plumb bob for centring. **Setting up** means levelling, centring and **orientation** — keeping the board parallel to its position at the first station. Orientation by **back-sighting** is more reliable than by trough compass.

Method	What is done	Used for
Radiation	Rays drawn from one station to all points; distances measured and scaled off	Small areas, all points visible from one station
Intersection	Rays drawn to the same point from two stations at the ends of a measured base; the crossing fixes the point	Distant or inaccessible points (across a river)
Traversing	Table moved from station to station, like a compass traverse	Narrow strips — roads, rivers
Resection	Finding on the sheet the position of the station occupied by the table, by sighting known points	Fixing new instrument stations

Resection problems. Two-point problem: the station is found from two plotted points. **Three-point problem:** from three plotted points — solved by the mechanical (tracing paper) method, the graphical (Bessel's) method or by trial and error (**Lehmann's rules**, which shrink the **triangle of error** to a point). The solution fails when the station lies on the circle passing through the three points (the **great circle**).

Photogrammetry



Scale = f / (H − h)

f = focal length of the camera
H = flying height above datum
h = elevation of the ground

2023 Q137: 0.2 / (2700 − 200)
= 1 in 12,500

Relief displacement d = r h / H′
r = radial distance of the image,
H′ = flying height above the base of the object

Higher ground is closer to the camera, so it is photographed at a larger scale. That is why one photograph has no single exact scale.

Aerial photographs — key facts

- **Vertical photograph:** camera axis vertical (tilt under about 3°). **Oblique:** axis deliberately tilted — high oblique shows the horizon, low oblique does not.
- **Principal point** — foot of the perpendicular from the lens on the photograph. **Nadir (plumb) point** — point vertically below the lens. On a truly vertical photograph they coincide.
- **Relief displacement** is radial from the nadir point: tops of tall objects are displaced outwards. Height of an object h = d H′ / r.
- **Overlap:** longitudinal (forward) about **60 %** between successive photographs of a strip, so that every point appears on two photographs and can be seen in 3-D; side (lateral) about **30 %** between adjacent strips.

- **Flight planning:** ground side covered by one photograph = photo size ÷ scale. Net new ground per photograph = (1 − P_l) L × (1 − P_w) W. Number of photographs = total area ÷ net area.
- 2023 Q139 worked: 0.2 m × 10,000 = 2 km each way → 4 km² per photograph; net = (0.4 × 2) × (0.6 × 2) = 0.96 km²; 192 / 0.96 = 200 photographs.
- **Stereoscopy:** an overlapping pair viewed under a stereoscope gives a 3-D model; heights are found from the difference in **parallax** measured with a parallax bar.

Map substitutes

When a proper line map is not available or would take too long, photographic products are used in its place. These are called **map substitutes**.

Map substitute	What it is
Mosaic	Several overlapping aerial photographs joined into one continuous picture. **Uncontrolled** — simply matched by detail; **controlled** — photographs rectified and fitted to ground control points; semi-controlled — in between
Photomap	A mosaic or single photograph with grid lines, place names, a scale and a north arrow added
Orthophoto / orthophotomap	A photograph from which tilt and relief displacement have been removed, so it has a uniform scale like a map
Photo index	A small-scale assembly of prints showing which photograph covers which area

4. Everyday analogy

Take a photograph of a street from a balcony: the nearby wall looks larger than the far end of the street. In an aerial photograph hilltops are the "nearby wall" — larger scale and pushed outward from the centre.

5. Confusing areas and exam traps

- Use **H − h**, not H, in the scale formula; and keep f and H in the same unit.
- In flight planning, first find the ground side of one photograph; do not multiply the two overlaps together directly.
- Intersection needs a measured base line; radiation needs every distance measured.
- Lehmann's rules belong to the trial-and-error solution of the three-point problem.

6. Connections

Stereo photographs are a main source of DEMs and contours (Topic 6). Satellite images extend the same idea from aircraft to space (Topic 10).

Memory code: “R-I-T-R” and “60 forward, 30 side”

R-I-T-R	Plane-table methods: Radiation, Intersection, Traversing, Resection
60 forward, 30 side	Longitudinal overlap 60 %, side overlap 30 %
f over H minus h	Scale of a vertical photograph = f / (H − h)

Self-recall (close the notes, answer aloud, then check)

1. Which plane-table method locates the station occupied by the table?
   Resection (two-point and three-point problems).
2. Why is 60 % forward overlap given?
   So that every ground point appears on at least two successive photographs, which is needed for stereoscopic (3-D) viewing.
3. What is an orthophoto?
   A photograph corrected for tilt and relief displacement, giving a uniform scale like a map.

Quick check: 5 questions

1. Vertical photographs are taken with a camera of focal length 150 mm from a flying height of 3200 m above datum. If the ground is 200 m above datum, the scale of the photographs is

   1. 1 in 21,333
   2. 1 in 20,000
   3. 1 in 15,000
   4. 1 in 30,000

1. Uses H instead of H − h.
2. Correct. Scale = 0.15/(3200 − 200) = 1/20,000.
3. Would need H − h = 2250 m.
4. Would need H − h = 4500 m.

2. An area of 112 km² is to be photographed at a scale of 1 : 10,000 with photographs of 20 cm × 20 cm, 60 % longitudinal overlap and 30 % side overlap. The number of photographs required is

   1. 28
   2. 70
   3. 100
   4. 280

1. Ignores both overlaps (112/4).
2. Allows only for the longitudinal overlap.
3. Correct. Ground side = 2 km; net area = (0.4 × 2) × (0.7 × 2) = 0.8 × 1.4 = 1.12 km²; 112/1.12 = 100.
4. Takes the net area as 0.4 km².

3. On a vertical photograph taken from 2000 m above the base of a tower, the top of the tower is imaged 80 mm from the principal point. If the tower is 50 m high, its relief displacement is

   1. 2 mm
   2. 1 mm
   3. 4 mm
   4. 8 mm

1. Correct. d = r h / H = 80 × 50/2000 = 2 mm.
2. Half the correct value.
3. Double the correct value.
4. Four times the correct value.

4. Lehmann's rules are used in plane tabling for

   1. orientation by trough compass
   2. the trial-and-error solution of the three-point problem
   3. the intersection method
   4. levelling the table

1. A trough compass needs no rules of this kind.
2. Correct. They guide the choice of the trial point so that the triangle of error shrinks to a point.
3. Intersection simply uses rays from two stations.
4. Levelling is done with a spirit level.

5. Consider the following statements:

   I. Orientation of a plane table by back-sighting is more reliable than by trough compass.
   II. The intersection method needs a measured base line.
   III. On a truly vertical photograph, relief displacement is radial from the principal point.
   IV. A mosaic is an assembly of overlapping aerial photographs.

   Which of the above statements are correct?

   1. I and II only
   2. I, II and III only
   3. II, III and IV only
   4. I, II, III and IV

1. Leaves out III and IV, which are true.
2. Leaves out IV.
3. Leaves out I.
4. Correct. All four are true. On a truly vertical photograph the nadir point and the principal point coincide.

Topic 10: Remote Sensing, GPS and Field Astronomy

Syllabus line: *Remote sensing concepts; concept of global positioning system; field astronomy.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 2 questions

- Match regions of the electromagnetic spectrum with wavelengths: visible 0.4–0.7 µm; near and mid infrared 0.7–3 µm; thermal infrared 3–14 µm; microwave 1 mm–1 m (Q135).
- Spectral resolution indicates the **wavelengths** to which the sensor is sensitive (Q138).

1. Basic idea

Remote sensing is collecting information about an object without touching it — in practice, recording the electromagnetic energy reflected or emitted by the earth's surface from an aircraft or satellite. **GPS** gives the position of a receiver anywhere on earth from satellite signals. **Field astronomy** uses the sun and stars to find true north, latitude, longitude and time.

A. Remote sensing

2. Why it is needed

A satellite sees a whole district in one image and returns every few days. That makes it the fastest way to map land use, floods, reservoirs, crops, forests and urban growth, and to choose routes for roads and canals.

3. How it works

Stages: energy source (sun) → passage through the atmosphere → interaction with the target → recording by the sensor → transmission and processing → interpretation → application.



Reflected sunlight is recorded in the visible and near/mid infrared bands; heat emitted by the ground in the thermal band; radar works in the microwave band.

Resolution	Meaning	Example
Spatial	Size of the smallest object that can be distinguished — the ground size of one pixel	LISS-IV: about 5.8 m; Cartosat: 2.5 m and finer
Spectral	The wavelengths to which the sensor is sensitive — number and width of its bands	Multispectral: a few broad bands; hyperspectral: hundreds of narrow bands
Radiometric	Sensitivity to small differences in energy — number of grey levels (bits)	8-bit = 256 levels; 10-bit = 1024 levels
Temporal	How often the same area is imaged again (revisit time)	A few days to a few weeks

More remote-sensing facts

- **Passive** sensors record sunlight reflected or heat emitted by the earth (cameras, multispectral scanners). **Active** sensors send their own energy and record the return (radar, LiDAR) — they work at night and through cloud.
- **Spectral signature:** each surface reflects differently in each band. Healthy vegetation absorbs red and reflects strongly in the near infrared; clear water absorbs near infrared and looks dark.
- **NDVI** = (NIR − Red)/(NIR + Red) — a measure of green vegetation.
- **Standard false colour composite:** near infrared shown as red, red as green, green as blue — healthy vegetation appears red.
- **Scattering:** Rayleigh (particles much smaller than the wavelength; makes the sky blue), Mie (particles about the size of the wavelength), non-selective (larger particles; makes clouds white). **Atmospheric windows** are the bands that pass through the atmosphere with little loss.
- **Orbits:** sun-synchronous near-polar orbits at about 700–900 km for earth observation (IRS, Resourcesat, Cartosat); geostationary orbit at about 36,000 km for weather and communication (INSAT).
- **Elements of visual interpretation:** tone, texture, shape, size, pattern, shadow, site and association.

B. Global Positioning System

GPS — the concept

- **Three segments:** space segment (the satellites), control segment (ground stations that track and correct them) and user segment (receivers).
- **Constellation:** nominally 24 satellites in 6 orbital planes inclined at 55°, at an altitude of about 20,200 km, each completing an orbit in about 12 hours.
- **Principle:** the receiver measures the travel time of the signal from each satellite, converts it into a distance (range), and fixes its position by **trilateration**.
- **At least 4 satellites** are needed: three for the position (X, Y, Z) and one more to remove the error of the receiver's clock.
- Signals are broadcast on the L-band (L1 and L2 carriers). Positions are given on the **WGS-84** datum.
- **Errors:** satellite clock and orbit (ephemeris) errors, ionospheric and tropospheric delay, multipath (reflected signals), receiver noise, and poor satellite geometry (high DOP).
- **DGPS:** a reference receiver on a known point computes the errors and sends corrections to nearby receivers, improving accuracy from several metres to a metre or better. RTK gives centimetre accuracy.
- Other systems: GLONASS (Russia), Galileo (Europe), BeiDou (China), and India's regional system **NavIC (IRNSS)**; GAGAN is India's augmentation system for aircraft.

C. Field astronomy

Term	Meaning
Celestial sphere	Imaginary sphere of infinite radius, centred on the observer, on which the stars appear fixed
Zenith and nadir	Points vertically above and below the observer
Celestial equator	Great circle where the plane of the earth's equator cuts the celestial sphere
Observer's meridian	Great circle through the zenith and the celestial poles
Latitude (φ) / co-latitude	Angular distance of the place from the equator / 90° − φ
Declination (δ) / polar distance	Angular distance of a star from the celestial equator / 90° − δ
Altitude (α) / zenith distance	Angle of the star above the horizon / 90° − α
Azimuth	Horizontal angle between the observer's meridian and the vertical circle through the star
Hour angle	Angle at the pole between the observer's meridian and the declination circle of the star, measured westward
Right ascension	Angle along the celestial equator from the First Point of Aries eastward to the star's declination circle

Rules to remember

- **Altitude of the celestial pole = latitude of the observer.**
- **Astronomical triangle:** formed by the pole, the zenith and the star; its sides are the co-latitude, the zenith distance (co-altitude) and the polar distance (co-declination).
- A star is **circumpolar** (never sets) when its declination is greater than the co-latitude: $\delta > 90° - \varphi$.
- Upper culmination of a star north of the zenith: zenith distance = $\delta - \varphi$. Lower culmination: altitude = $\varphi + \delta - 90°$.
- **Time:** 360° of longitude = 24 hours → **15° = 1 hour, 1° = 4 minutes**. Places to the east have later local time. Indian Standard Time is the local mean time of longitude 82°30′ E = GMT + 5 h 30 min.
- A **sidereal day** (23 h 56 min 4 s of mean solar time) is about 4 minutes **shorter** than a mean solar day.
- **Equation of time** = apparent solar time − mean solar time; it varies through the year up to about 16 minutes and is zero four times a year.
- Corrections to an observed altitude: refraction (subtract), parallax (add), semi-diameter (for the sun), dip (at sea).

4. Everyday analogy

Remote sensing is like a doctor reading an X-ray instead of opening the body. GPS is like finding where you are in a dark hall by listening to four friends who shout the exact time from known corners — the delay of each voice tells you how far each friend is.

5. Confusing areas and exam traps

- **Spatial** = detail; **spectral** = wavelengths; **radiometric** = energy sensitivity; **temporal** = revisit frequency. The 2023 options were exactly these four.
- GPS needs **four** satellites, not three, for a 3-D fix.
- GPS satellites are at about 20,200 km — not geostationary (36,000 km).
- A sidereal day is shorter than a solar day; east is ahead in time.

6. Connections

Satellite images are processed like aerial photographs (Topic 9) and yield DEMs (Topic 6). GPS coordinates replace triangulation control (Topic 8). The true meridian found by astronomy is the reference for bearings (Topic 3).

Memory code: "S-S-R-T" and "4 – 6 – 55 – 12 – 20"

S-S-R-T	Resolutions: Spatial (size), Spectral (wavelength), Radiometric (energy levels), Temporal (time)
4 – 6 – 55 – 12 – 20	GPS: 4 satellites minimum, 6 orbital planes, 55° inclination, 12-hour period, 20,200 km altitude
Pole height = Place latitude	Altitude of the celestial pole equals the latitude of the observer

Self-recall (close the notes, answer aloud, then check)

1. Give the wavelength range of the visible band and of the thermal infrared band.
   Visible 0.4–0.7 µm; thermal infrared 3–14 µm.
2. Why does GPS need a fourth satellite?
   Three ranges fix X, Y and Z; the fourth removes the receiver clock error.
3. How much time corresponds to 1° of longitude?
   4 minutes (15° = 1 hour).

Quick check: 5 questions

1. Match List-I (type of resolution) with List-II (what it describes):

List-I	List-II
A. Spatial	1. Revisit time of the satellite
B. Spectral	2. Number and width of wavelength bands
C. Radiometric	3. Ground size of one pixel
D. Temporal	4. Number of grey levels (bits)

1. A-2, B-3, C-4, D-1
2. A-3, B-2, C-4, D-1
3. A-3, B-4, C-2, D-1
4. A-1, B-2, C-4, D-3

1. Swaps spatial and spectral.
2. Correct. Spatial → ground size of a pixel; spectral → wavelength bands; radiometric → grey levels; temporal → revisit time.
3. Swaps spectral and radiometric.
4. Swaps spatial and temporal.

2. The minimum number of GPS satellites needed to find the three-dimensional position of a receiver is

1. 2
2. 3
3. 4
4. 6

1. Two ranges cannot fix a point in space.
2. Three would be enough only if the receiver clock were perfect.
3. Correct. Three for X, Y, Z and one for the receiver clock error.
4. More satellites improve accuracy, but four is the minimum.

3. When Indian Standard Time is 10:00, the local mean time at a place 7°30′ east of the standard meridian is

1. 10:30
2. 09:30
3. 10:07
4. 09:52

1. Correct. 7°30′ × 4 min per degree = 30 min; places to the east are ahead, so 10:30.
2. East is ahead, not behind.
3. Uses 1 minute per degree.
4. Uses 1 minute per degree and the wrong sign.

4. Consider the following statements:

I. The altitude of the celestial pole is equal to the latitude of the observer.
II. Declination is measured from the celestial equator.
III. A sidereal day is longer than a mean solar day.
IV. Equation of time = apparent solar time − mean solar time.

Which of the above statements are correct?

1. I and II only
2. I, III and IV only
3. II, III and IV only
4. I, II and IV only

1. Leaves out IV, which is true.
2. III is false.
3. III is false.
4. Correct. A sidereal day is about 4 minutes shorter than a mean solar day, so III is false.

5. **Assertion (A):** In a standard false colour composite, healthy vegetation appears red.
Reason (R): Healthy vegetation reflects strongly in the near infrared, and the near-infrared band is displayed in red.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. Correct. Both are true and R explains A.
2. R is the direct reason.
3. R is true.
4. A is true.

Notebook page — write these from memory

Close this page. In your notebook, write the items below in this order without looking. Then open the page and correct yourself in a different colour. Whatever you could not write is your revision list for Sunday.

Topic 1

1. Definition of surveying; the two principles
2. Plane vs geodetic; classification table
3. RF, types of scale, least count of a vernier, shrunk-scale rule
4. Three kinds of error and how each grows with length

Topic 2

1. Types of chain and tape with lengths
2. Base, check and tie lines; offsets; well-conditioned triangle
3. Wrong-chain formulas for length, area, volume
4. Five tape corrections with signs; normal tension
5. Cumulative and compensating errors; the three kinds of obstacle; the river sketch

Topic 3

1. Four meridians
2. Quadrant sketch with the four WCB → RB rules; FB and BB
3. Prismatic vs surveyor's compass table
4. Declination rule; isogonic, agonic, isoclinic, aclinic
5. Local attraction: test and correction

Topic 4

1. Types of theodolite; fundamental lines and their relations; temporary adjustments
2. Repetition vs reiteration; which error is removed by what
3. Latitude–departure sketch with formulas; closing error
4. Angle checks; Bowditch and transit rules; consecutive vs independent coordinates

Topic 5

1. Terms: datum, RL, BM, BS, IS, FS, change point, HI
2. Levelling sketch; HI method and rise-and-fall method with checks
3. 0.0785 d^2, 0.0112 d^2, 0.0673 d^2, 3.855 $\sqrt{h}$
4. Types of levelling (one line each); sensitivity of the level tube

Topic 6

1. Contour, contour interval, horizontal equivalent
2. Four sketches: hill, depression, valley, ridge
3. Characteristics of contours
4. Direct and indirect methods; DEM, TIN, DSM, DTM and sources

Topic 7

1. Stadia sketch; D = ks + C; anallactic lens
2. Fixed-hair vs movable-hair; tangential; subtense bar
3. Inclined-sight formulas for D and V
4. EDM principle and three types; what a total station measures and computes

Topic 8

1. Principle of triangulation; table of the three orders
2. Three figures with their merits; well-conditioned triangle; signals; satellite station
3. Laws of accidental error; most probable value; least squares; weight rules
4. Station and figure adjustment; distribution by weights

Topic 9

1. Plane-table accessories and orientation
2. R-I-T-R with one use each; two-point and three-point problems; Lehmann's rules
3. Aerial-photo sketch; scale = f/(H − h); relief displacement
4. Overlaps; flight-planning steps (the 2023 example); map substitutes

Topic 10

1. Stages of remote sensing; spectrum strip with wavelengths
2. Four resolutions; passive vs active; FCC and NDVI; orbits
3. GPS: segments, constellation numbers, why four satellites, errors, DGPS
4. Astronomy terms; pole altitude = latitude; astronomical triangle; time rules

References

- APPSC Notification No. 27/2026 (Assistant Executive Engineers) — syllabus of Paper 3 Civil, item 9 (Surveying).
- APPSC AEE Paper III (Civil Engineering), held on 22 Aug 2023, Shift 2 — official paper with key. All "Asked in APPSC AEE 2023" notes refer to this paper.
- B.C. Punmia, *Surveying*, Volumes 1 to 3 — formula forms, sign conventions and classification values follow this book.
- Photos: Wikimedia Commons, free licences; each photo links to its source page.
- Quiz questions are original practice questions (Sulabham), not previous-year questions.

Facts verified on 11 Nov 2026.

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